

AEGIS-CARE

A Privacy-Bounded Memory Recompiler for Recovering Poisoned Clinical AI Agents

*Latent dependency discovery, local counterfactual confirmation,
clean-room replay, and resurrection-resistant repair under incomplete provenance*

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CORE QUESTION

When one clinical AI memory is found to be wrong, can the system discover, repair, and verify everything that inherited the error without opening every patient record?

Proposal at a Glance

MENTOR DECISION REQUEST

Approve a focused capstone study of post-breach recovery in a simulated, role-separated EHR assistant: three agents, one primary open-weight model, one CARE recovery algorithm, one wrong-patient propagation family, and an ablation-first evaluation against deletion, reset, explicit-lineage, and centralized-oracle baselines.

AEGIS-Care studies a failure that existing guardrails do not fully address. A clinical record assistant may store an incorrect patient association or unauthorized record fragment, summarize it into later memory, and reuse the derived state after the original source is corrected. Deleting the visible seed may therefore leave authentic-looking handovers, summaries, and retrieval cues active. A full reset removes the error but also destroys useful memory. Central forensic reconstruction may recover accurately but require access to patient content that the investigator is not authorized to read.

The proposed system treats agent memory as a versioned dataflow program. A known compromised seed invalidates its downstream artifacts. The CARE loop finds candidate descendants, confirms influence through local counterfactual replay, recompiles repairable memories from trusted FHIR sources, and prevents old influence from being reintroduced. A narrow latent recovery mesh is used only to discover candidates when explicit lineage is missing; full hidden states and KV caches are never exchanged.

Decision	Proposal commitment
Domain	Simulated clinical record management, not diagnosis or treatment.
Primary incident	Propagated wrong-patient association across registration, nursing handover, and clinical summary memories.
Technical core	CARE: Candidate discovery, Attribution, Recompilation, Enforcement.
Privacy boundary	Raw EHR content remains inside role-authorized local runtimes; the coordinator receives scoped recovery evidence.
Latent component	Compressed, receiver-scoped dependency sketches for candidate discovery only.
Primary proof	Lower residual wrong-patient/unauthorized behavior than privacy-respecting baselines, with greater clean-memory retention than full reset.
Data	MedAgentBench-derived FHIR environment plus EICU-AC-style role policies and synthetic incident overlays.
Claim boundary	Empirical recovery and leakage trade-offs; no clinical deployment, HIPAA certification, formal differential privacy, or universal guarantee.

Plain-language value: if an AI remembers the wrong patient, AEGIS-Care does not merely erase one note or wipe everything. It finds where that memory spread, repairs what can be safely rebuilt, locks what cannot, and shows that unrelated records remained untouched.

Abstract

Persistent memory enables clinical AI agents to reuse patient identifiers, prior retrievals, handover summaries, and workflow knowledge across sessions. The same persistence can convert one corrupted or unauthorized memory write into a chain of derived artifacts that continue to affect future record-management tasks after the original source is corrected. Existing work demonstrates long-term-memory poisoning, provenance-aware action gating, role-based access control, and deterministic replay, but does not directly solve privacy-bounded post-incident reconstruction when lineage is incomplete and the affected memories reside behind different clinical roles. This proposal introduces AEGIS-Care, a versioned memory-recompilation system for simulated clinical agents. Its CARE loop combines explicit lineage with receiver-scoped latent dependency sketches to generate candidate descendants; performs counterfactual replay inside each authorized agent boundary to confirm material influence; reconstructs affected memories from trusted FHIR resources; and applies tombstones plus a resurrection firewall to prevent stale influence from returning. Evaluation will overlay controlled wrong-patient and access-scope contamination on MedAgentBench workflows under complete, random-loss, and targeted-loss provenance. AEGIS-Care will be compared with no recovery, seed deletion, full reset, explicit-lineage quarantine, explicit-lineage replay, and a centralized raw-content oracle. Primary outcomes are residual wrong-patient or unauthorized behavior, contaminated-descendant recall and precision, clean-state retention, repaired task success, unauthorized exposure, resurrection, and overhead. The study uses deidentified/synthetic benchmark records only and excludes diagnosis, treatment recommendation, medication ordering, real EHR deployment, and formal compliance claims.

Proposal Roadmap

1. Introduction and clinical motivation
2. Problem statement and exact research gap
3. Aim, objectives, research questions, and hypotheses
4. Literature review and positioning
5. Proposed AEGIS-Care architecture and CARE algorithm
6. Formal model, threat model, privacy model, and invariants

7. Clinical benchmark, experimental design, metrics, and statistics
8. Functional scope, feasibility, technology stack, and timeline
9. Risks, ethics, expected contribution, publication plan, and conclusion
10. References and implementation appendix

1. Introduction and Clinical Motivation

Large language model agents differ from ordinary chatbots because they can interpret a high-level request, call tools, read and modify external systems, and retain state for later tasks. In healthcare, this could reduce documentation and information-retrieval burden. MedAgentBench operationalizes this setting through 300 clinician-written tasks across a FHIR-compliant virtual EHR containing 100 patient profiles and more than 700,000 data elements [4]. Its task families include patient lookup, laboratory and vital retrieval, aggregation, recording observations, and simulated orders. The benchmark also shows that strong models remain imperfect, which makes system-level reliability and recovery important rather than optional.

AEGIS-Care deliberately selects the lowest-risk, most defensible subset of this opportunity: patient-record identification, authorized information retrieval, handover summarization, data aggregation, and simulated documentation. It will not diagnose disease, recommend treatment, calculate medication doses for execution, or operate on a live EHR. The research question is not whether an LLM can practice medicine. It is whether a memory-enabled record assistant can recover safely when its durable state becomes wrong.

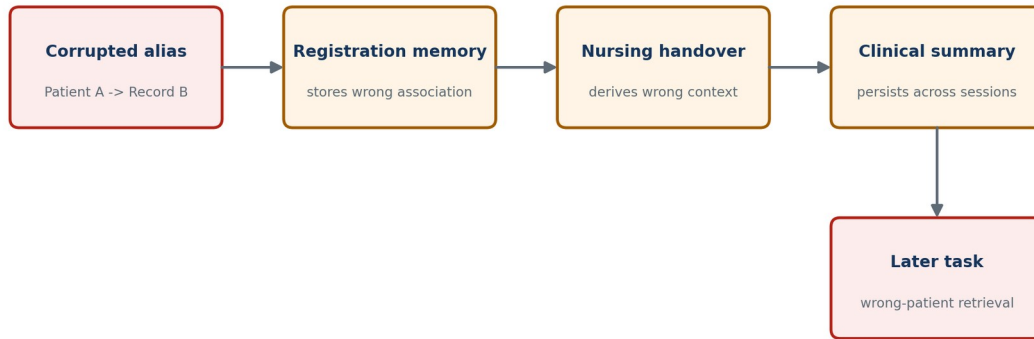
1.1 Why wrong-patient memory is the right tangible problem

Patient identification is a universal step in recording and accessing health information. The Joint Commission notes that wrong-patient errors can arise throughout health IT workflows, including information entered into the wrong chart, co-mingled records, duplicate charts, and tests assigned to the wrong patient [2]. WHO similarly identifies patient misidentification as a cause of serious adverse effects [1]. Copy-forward and copy-paste workflows can also propagate internal inconsistencies and wrong-chart documentation [3]. These established failure patterns provide a non-technical, socially meaningful analogue for propagated agent memory.

The proposed benchmark will not claim to reproduce hospital harm. It will reproduce the information-system mechanism in a controlled environment: a wrong or unauthorized patient association is written into agent memory; later agents derive handovers or summaries from it; the seed is subsequently

flagged; and recovery methods are evaluated on whether downstream state is repaired without exposing unrelated records.

One wrong durable association can become several authentic-looking memories



Deleting only the original alias does not remove the handover or summary that already inherited it.

Figure 1. Primary AEGIS-Care incident: a single corrupted association survives through derived clinical-agent memory.

1.2 Why ordinary deletion is insufficient

Persistent-agent attacks now demonstrate that untrusted material can be written once and influence behavior in later sessions [6-9]. MPBench identifies explicit writes, system-prompt-driven writes, compaction-driven writes, and experience-to-procedure promotion as distinct memory channels, and reports that more aggressive memory use increases exploitability [7]. In a clinical workflow, a wrong association may therefore survive as a summary, retrieval cue, cached procedure, or agent-authored note even after the original record is corrected.

RECOMPILER INSIGHT

A database rollback changes the source record. A memory recompiler also invalidates and rebuilds the semantic artifacts that were previously derived from that source.

2. Problem Statement and Exact Research Gap

Consider three role-separated assistants. The registration agent can resolve identity and appointments. The nursing agent can access observations and handover material. The clinical-summary agent can produce an authorized overview. During one session, a corrupted alias associates Patient A with Record B. That association is retrieved into a handover and later into a summary. When the original alias is

discovered, the recovery service knows at least one compromised seed but may not know every derived edge. It also lacks permission to centralize the nursing or clinical text needed for a naive forensic reconstruction.

CORE PROBLEM

Given a known compromised memory seed, incomplete provenance, and role-bounded evidence access, identify and safely reconstruct every materially affected descendant while preserving independent memory and preventing unauthorized disclosure or reinfection.

2.1 Why current approaches do not close the loop

Approach	What it solves	What remains
Prompt-injection guard	May block malicious input in the current interaction.	Does not repair durable derived state already written.
Role-based access control	Prevents an unauthorized role from directly reading protected fields.	Does not locate cross-session contamination behind authorized role boundaries.
Delete the flagged seed	Removes the known bad entry.	Descendant summaries and cached procedures may remain active.
Full reset	Provides a strong safety fallback.	Destroys legitimate memory and eliminates the benefit of persistence.
Provenance-only cascade	Works when all relevant edges were captured.	Misses descendants under lossy, redacted, or cross-agent lineage.
Central raw-content investigation	Can maximize evidence available to recovery.	Creates a privacy and governance problem by centralizing patient content.
Similarity-only recovery	Finds semantically related memories.	Confuses relatedness with derivation and may over-quarantine clean state.

2.2 Precise research gap

MemLineage shows that signed provenance and derivation-aware labels can prevent untrusted ancestry from authorizing sensitive actions [10]. Execution-lineage work shows that explicit artifact graphs and identity-based replay support partial recomputation after changes [11]. GuardAgent and EICU-AC show how clinical-agent actions can be checked against role-based access policies [5]. Latent-communication research shows that continuous representations can carry more information than text, while also introducing opacity, integrity, and privacy risks [13-15]. These research lines motivate the components, but they do not provide a combined, evaluated mechanism for post-incident recompilation across privacy-separated clinical-agent memories when explicit lineage is incomplete.

AEGIS-Care therefore does not claim that provenance, replay, access control, latent representations, or memory poisoning are individually new. Its research contribution is the recovery problem and the controlled algorithmic composition required to solve it: latent candidate discovery is never trusted as causality; local counterfactual replay confirms influence; clean-room recompilation restores utility; and version enforcement prevents resurrection. Each stage exists because the preceding stage has a measurable failure mode.

3. Aim, Objectives, Research Questions, and Hypotheses

3.1 Aim

To design and evaluate AEGIS-Care, a privacy-bounded memory-recompilation framework that recovers propagated wrong-patient and unauthorized-record contamination in role-separated clinical AI agents under incomplete provenance.

3.2 Objectives

1. Construct a reproducible, multi-session clinical record-assistant environment using a MedAgentBench-derived FHIR sandbox and three EICU-AC-inspired roles.
2. Represent durable agent state as versioned memory artifacts with explicit parent commitments, patient/role/purpose policy, replay recipe, and revocation status.
3. Design receiver-scoped latent recovery sketches that improve candidate-descendant discovery when explicit lineage edges are absent.
4. Confirm suspected influence through local counterfactual replay rather than treating semantic similarity as causal proof.
5. Recompile repairable memories from trusted FHIR sources, quarantine unreconstructable artifacts, and recursively update downstream state.
6. Prevent reinfection through tombstones, version checks, and a resurrection firewall applied to later retrievals and writes.
7. Evaluate safety, utility, privacy leakage, calibration, and overhead against realistic non-oracle and oracle baselines.
8. Produce an auditable benchmark overlay, implementation, ablation report, and submission-ready research paper.

3.3 Research questions and hypotheses

ID	Research question	Falsifiable hypothesis
RQ1	How quickly does recovery fail as provenance edges are removed?	Targeted loss of cross-role and semantic-derivation edges will harm provenance-only recovery more than random edge loss.
RQ2	Do scoped latent sketches improve descendant discovery?	At matched candidate budgets, explicit lineage plus sketches will recover more true descendants than explicit lineage alone.
RQ3	Can local counterfactual replay separate causal inheritance from mere semantic similarity?	Replay confirmation will improve precision and clean-state retention relative to sketch-only quarantine at comparable recall.
RQ4	Does recompilation restore more useful state than deletion, quarantine, or reset?	Clean-room reconstruction will retain higher post-recovery task success than full reset and quarantine-only baselines while maintaining low residual harm.
RQ5	What privacy is lost through the recovery channel?	Scoped sketches and signed verdicts will leak less protected attribute/membership information than raw text or unrestricted embeddings, at a measurable recovery cost.
RQ6	Does the resurrection firewall prevent re-entry?	Version and revocation checks will reduce reappearance of withdrawn influence across clean follow-up sessions.

3.4 Success is a frontier, not one accuracy number

A system can appear safe by deleting everything, private by revealing nothing, or useful by ignoring contamination. The primary analysis will therefore compare methods at matched clean-state retention and report a safety-utility-privacy frontier. The headline claim will be accepted only if AEGIS-Care lowers residual wrong-patient or unauthorized behavior beyond the strongest privacy-respecting non-oracle baseline while preserving materially more legitimate state than reset.

4. Literature Review and Positioning

The literature is organized by the function it contributes to the proposed system. Peer-reviewed sources are distinguished from recent 2026 preprints; the proposal does not make a priority claim based only on keyword absence.

Research line	Established result	Boundary used by AEGIS-Care
Clinical agent evaluation [4]	MedAgentBench provides clinician-written FHIR tasks, patient profiles, deterministic graders, and public code.	It evaluates task execution, not recovery after persistent-memory contamination.
Clinical access control [5]	GuardAgent/EICU-AC operationalizes physician, nursing, and administration access policies.	It guards actions but does not rebuild already derived memory across roles.
Memory poisoning [6-9]	AgentPoison, MemoryGraft, sleeper poisoning, and MPBench establish persistent cross-session compromise.	The main contribution is attack or prevention analysis rather than private descendant repair.

Research line	Established result	Boundary used by AEGIS-Care
Lineage enforcement [10]	MemLineage attaches signed provenance and derivation lineage and gates sensitive actions.	AEGIS-Care focuses on post-incident cleanup, incomplete edges, cross-role stores, and reconstruction.
Execution replay [11]	Artifact DAGs and identity-based replay preserve unaffected work while propagating changes.	Clinical privacy, poisoned-memory attribution, and missing lineage are outside its main scope.
Provenance-aware memory [12]	Tiered memory highlights that lossy summaries can omit decisive evidence and need verifiable sources.	Does not provide privacy-bounded post-breach recompilation.
Latent communication [13-15]	Embeddings, hidden states, and KV caches can transmit dense intermediate evidence.	Opacity and latent attacks make full latent exchange inappropriate; AEGIS uses only narrow recovery sketches.
Embedding leakage [16]	Text/hidden representations can support input reconstruction and attribute inference.	Sketches must be treated as potentially leaking and evaluated rather than called private by default.
Patient safety [1-3]	Misidentification and copied-forward inconsistencies are recognized clinical information risks.	AEGIS-Care studies a simulated information-system mechanism, not clinical outcome effectiveness.

4.1 The immediate comparator: MemLineage

MemLineage is the closest technical comparator because it records weighted derivation edges, binds provenance cryptographically, and prevents sensitive actions justified by untrusted ancestry [10]. AEGIS-Care should therefore not present lineage tracking as new. The proposed differentiation is narrower: recovery begins after a seed is confirmed; some lineage is unavailable; stores are separated by role; and the desired output is repaired state, not only blocked action. The evaluation includes a MemLineage-style explicit-lineage quarantine/gate baseline to test whether the additional candidate-confirm-recompile loop earns its complexity.

4.2 Why full latent mesh is rejected

A full latent mesh would exchange hidden states or KV caches as a general collaboration medium. Recent surveys identify model compatibility and auditability challenges [13]. Latent-attack studies show that adversarial behavior can be introduced through representation-level interventions [14], while KV-cache integrity work shows that visible commitments may fail to validate transported hidden state [15]. Full latent exchange would therefore add a second research project and weaken the privacy story.

LATENT DESIGN DECISION

AEGIS-Care uses latent information only as a compressed, receiver-scoped candidate-discovery sketch. It is never accepted as a trusted instruction, clinical fact, or proof of causality.

5. Proposed System: AEGIS-Care

5.1 System thesis

AEGIS-Care models persistent agent memory as a versioned program whose derived artifacts can be invalidated and incrementally recomputed. The system contains three role-specific agent runtimes, a FHIR source of truth, local memory vaults, and a recovery coordinator. Raw patient content remains in the local runtime authorized to process it. The coordinator orchestrates recovery using opaque commitments, policy-safe metadata, compressed sketches, and signed local verdicts.

AEGIS-Care: local records, minimal recovery evidence, central coordination

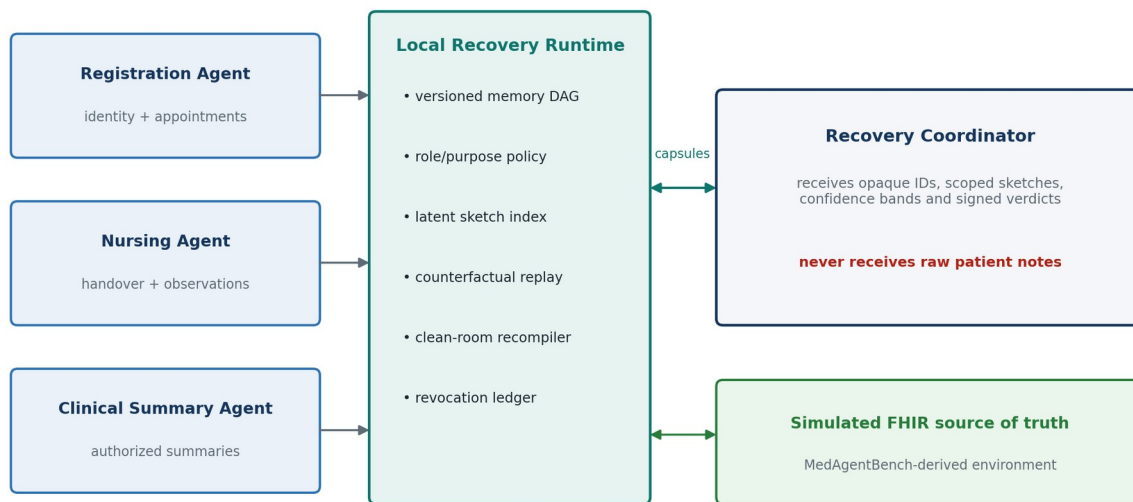


Figure 2. Proposed architecture. Recovery coordination is centralized; patient content and replay remain role-local.

5.2 Role-separated clinical agents

Agent	Authorized functions	Persistent memory examples	Excluded functions
Registration / administration	Patient lookup, demographic matching, appointments.	Resolved identity hints, scheduling preferences, prior lookup strategies.	Clinical notes, diagnoses, detailed labs.
Nursing handover	Authorized observations, vitals, documentation, handover summaries.	Shift handovers, observation summaries, workflow reminders.	Unrelated patients, physician-only restricted fields.
Clinical summary	Authorized cross-record summarization for a declared patient/purpose.	Problem-oriented summaries and retrieval plans.	Autonomous diagnosis, treatment, medication execution.

5.3 Versioned memory object

Field	Purpose
memory_id, version, owner	Stable identity for rollback, replacement, and role ownership.
content_ref	Pointer to encrypted/local content; never included in a recovery capsule.
explicit_parent_commitments	Known source and memory dependencies represented by hashes/opaque IDs.
patient_scope, role_scope, purpose	Deterministic authorization boundary for retrieval and forensic use.
write_context_sketch	Compressed local representation used to shortlist hidden descendants.
replay_recipe	Model/prompt/tool version, input commitments, schema, and deterministic task predicate needed for reconstruction.
status	active, suspected, quarantined, superseded, repaired, or tombstoned.
signature and timestamps	Integrity, attribution, temporal ordering, and auditability.

5.4 Scoped recovery capsule

When a seed is revoked, the coordinator does not broadcast its text. The owning runtime creates a receiver-specific capsule containing a seed commitment, recovery purpose, recipient, expiry, nonce, policy-safe type/time information, a compressed latent sketch, and a signature. Patient names, raw MRNs, notes, laboratory values, and hidden/KV states are excluded. Patient identity may be represented by a receiver-specific keyed token so repeated recovery events cannot be globally linked by an honest-but-curious coordinator.

Capsule field	Security/privacy function
seed_commitment	Binds the incident to a known version without exposing content.
recipient + purpose + expiry	Prevents general reuse of forensic evidence.
receiver-scoped patient token	Allows local matching while reducing cross-recipient linkability.
quantized latent sketch	Generates candidates under missing explicit lineage.
type/time band	Prunes impossible relationships without exact timestamps.
signature + nonce	Detects tampering and replay of a stale capsule.

5.5 The CARE loop

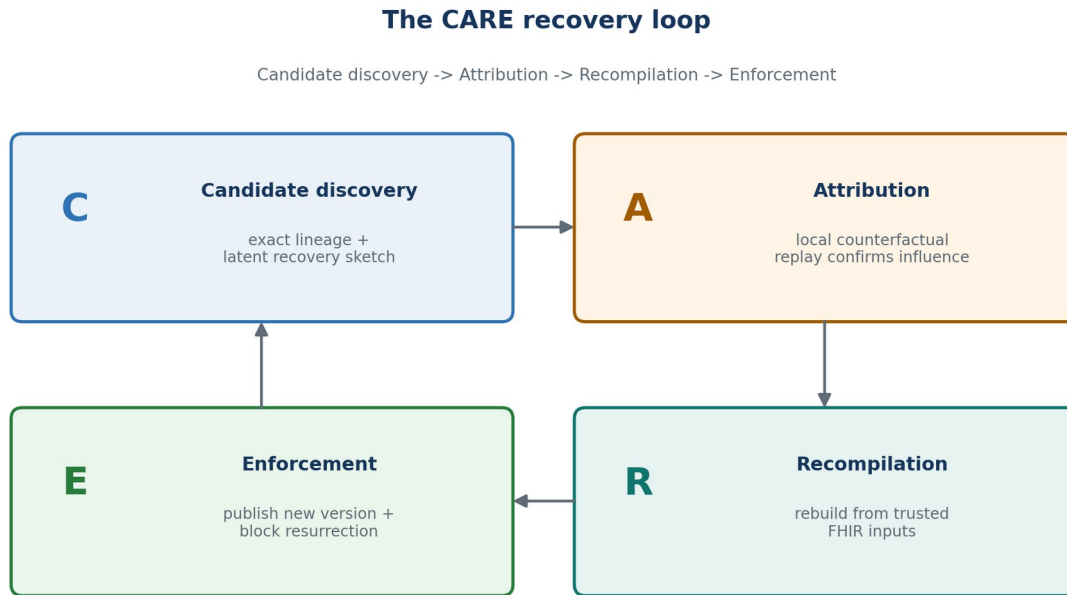


Figure 3. CARE is one recovery loop: Candidate discovery, Attribution, Recompilation, and Enforcement.

5.5.1 C - Candidate discovery

Each local runtime first traverses exact parent commitments. If a lineage edge is missing or withheld, it searches the local write-context sketch index using the receiver-scoped seed sketch, then applies local temporal, artifact-type, patient-token, and workflow compatibility filters. The latent component produces a ranked candidate set only. It does not quarantine or repair memory by itself.

5.5.2 A - Attribution through local counterfactual replay

For each candidate child v , the owning runtime replays its creation without the suspected seed or matching context item. Influence is confirmed when the replay materially changes a protected field, selected patient, FHIR resource, structured fact, or downstream action. The coordinator receives a signed influence band and recommended disposition, not the original or counterfactual clinical text.

5.5.3 R - Clean-room recompilation

A confirmed descendant is rebuilt from trusted, currently authorized FHIR resources and unaffected parent memories. The replay recipe pins the tool schema, prompt version, output schema, and task predicate. If required support is unavailable, contradictory, or outside the role's access rights, the item is quarantined for human review instead of being guessed. Repaired versions receive new IDs/versions and explicit provenance; descendants are then processed recursively.

5.5.4 E - Version enforcement and resurrection prevention

Superseded and poisoned versions remain as signed tombstones in the audit log but are excluded from retrieval. Future writes and retrievals are checked against revocation commitments and candidate sketches. A write that appears to reintroduce withdrawn influence is blocked, locally replayed, or escalated. This converts recovery from a one-time cleanup into an enforced state transition.

UNIFIED CONTRIBUTION

The latent sketch finds what may have inherited the error; counterfactual replay tests whether it actually did; recompilation restores useful state; enforcement keeps the repaired system repaired.

6. Formal Problem Formulation and Algorithm

6.1 Memory and policy graph

Let $G = (V, E)$ be the private ground-truth memory graph. Each node v in V is a versioned artifact owned by one role-specific runtime. An edge (u, v) in E means that u materially influenced the creation of v . The operational system observes E_{obs} , a possibly incomplete subset of E . Every node carries an access policy $\pi(v)$, a replay recipe $\rho(v)$, and a state $q(v)$ in $\{\text{active, suspected, quarantined, superseded, repaired, tombstoned}\}$. The incident begins with a known compromised seed set S .

A node is contaminated if it lies on a material-influence path from S and cannot be independently reproduced from trusted support. This definition avoids treating every semantically related node as contaminated and allows a derived memory to be cleared when clean-room replay produces an equivalent, policy-valid result.

6.2 Candidate scoring

Candidate scoring is performed locally so protected fields remain available to the owner but not the coordinator. For seed s and local memory v , the runtime computes:

$$\text{score}(s, v) = a * \text{explicit_edge}(s, v) + b * \text{sim}(z_s, z_v) + c * \text{compat}(\text{time}, \text{type}, \text{patient}, \text{workflow})$$

Here z is a quantized receiver-scoped sketch and compat is evaluated inside the local policy boundary. Threshold τ_c controls the candidate budget. Exact edges bypass the similarity threshold.

6.3 Counterfactual influence

Let v_minus_s be the replayed version of v after excluding the suspected influence. A structured influence function combines semantic change with changes to safety-critical task predicates:

$$I(s \rightarrow v) = w1 * \text{semantic_delta}(v, v_minus_s) + w2 * \text{patient_change} + w3 * \text{resource_or_action_change}$$

The edge is confirmed when I exceeds τ_i or when an exact deterministic predicate changes, such as the selected patient resource. Thresholds are selected on development scenarios at a frozen clean-state-retention floor.

6.4 Recovery objective

The algorithm chooses quarantine and repair sets that jointly minimize residual harm, unnecessary deletion, unauthorized disclosure, and operational cost:

$$J = \lambda_s * RWH + \lambda_u * (1 - BSR) + \lambda_p * UER + \lambda_r * DRR + \lambda_c * \text{Cost}$$

RWH is residual wrong-patient/unauthorized harm, BSR is benign-state retention, UER is unauthorized exposure, DRR is deletion-resurrection rate, and Cost is latency/model-call/capsule overhead. The optimization is subject to the hard constraint that the coordinator cannot receive raw clinical content outside its declared authorization.

6.5 Algorithm sketch

INPUT: revoked seed set S , observed lineage E_{obs} , role policies, local memory stores

1. Mark each seed non-servable and issue signed, receiver-scoped recovery capsules.
2. In each local runtime, collect exact descendants and latent-sketch candidates.
3. For every non-exact candidate, replay locally without the suspected influence.
4. If influence is not confirmed, retain the memory and record the negative verdict.
5. If confirmed and trusted support is sufficient, rebuild and verify a new version.
6. If confirmed but reconstruction is unsafe or impossible, quarantine and escalate.
7. Treat each confirmed descendant as a new frontier node; repeat until closure.
8. Commit tombstones, repaired versions, signed verdicts, and revocation fingerprints.
9. Run clean follow-up tasks plus resurrection probes before returning to service.

OUTPUT: repaired graph view, quarantined set, audit certificate, residual-risk report

6.6 Termination and safety properties

- Monotone incident frontier: within one recovery transaction, an artifact can move only toward suspected, quarantined, superseded, or repaired state; it cannot silently return to the original active version.
- Finite termination: with a finite version snapshot and no new writes during the barrier phase, each artifact is processed at most once per incident version; the loop terminates after the reachable candidate frontier is exhausted.
- No raw-content centralization: all content-dependent matching, replay, and verification occurs in the owning runtime.
- Fail-closed reconstruction: absence of trusted support produces quarantine or review, not speculative repair.
- Closed publication: a repaired artifact becomes retrievable only after policy, schema, identity, and task-predicate checks pass.

7. Threat Model, Privacy Model, and Invariants

Dimension	In scope	Out of scope
Failure source	Intentional poisoned memory write or accidental wrong-patient/Unauthorized association introduced through a legitimate workflow.	Model-weight compromise, OS compromise, stolen signing keys.
Incident knowledge	At least one compromised seed is confirmed before recovery begins.	End-to-end incident detection as the primary research problem.
Propagation	Summarization, compaction, retrieval cues, handover derivation, and procedure promotion across sessions.	Multimodal records and physical device control.
Privacy adversary	Honest-but-curious coordinator or unauthorized role attempting inference from allowed capsules, metadata, and query responses.	Collusion of all local runtimes or hardware side channels.
Trust base	Local policy engine, FHIR sandbox, replay runtime, signing keys, and ground-truth benchmark instrumentation.	A malicious local vault that fabricates replay evidence.
Guarantees	Deterministic scope enforcement, integrity-protected messages, empirical leakage and recovery measurements.	Formal differential privacy, zero-knowledge proof, FHE, TEE attestation, or legal compliance certification.

7.1 Core invariants

- A confirmed seed and every confirmed contaminated descendant are non-servable during recovery.
- No repaired memory may cite a tombstoned version as trusted support.

- No role may receive a raw record, sketch, or verdict outside the request's patient, role, purpose, recipient, and expiry policy.
- Latent similarity alone cannot trigger destructive repair; exact lineage or counterfactual confirmation is required.
- A missing replay recipe, missing trusted source, or policy conflict causes quarantine or human review.
- Recovery completion requires safety follow-up tasks, privacy checks, and resurrection probes.

7.2 Privacy measurement

The phrase privacy-bounded is used deliberately. A compact sketch is not inherently private; embedding inversion research shows that representations may reveal source text or attributes [16]. The system guarantees only that raw content is not intentionally exported through the defined recovery interface and that deterministic policy checks are applied. Residual leakage is measured empirically through protected-attribute prediction, membership inference, repeated-query/linkability probes, and exact logging of fields released to the coordinator.

8. Clinical Benchmark and Data Construction

8.1 Base environment

The implementation will reuse the public MedAgentBench FHIR environment and task schemas [4, 18]. The benchmark contains deidentified and jittered clinical records, with synthetic identifiers and demographic fields. AEGIS-Care will select read-heavy and low-risk tasks from patient information retrieval, laboratory/vital retrieval, data aggregation, and simulated documentation. Medication ordering, referral execution, and autonomous treatment are excluded from the primary experiment.

Role policies will be adapted from EICU-AC's physician, nursing, and general-administration structure [5, 19]. The project will implement deterministic policy checks over FHIR resource type, field category, patient token, role, purpose, and operation. A small synthetic overlay adds versioned agent memory, propagation traces, replay recipes, and contamination labels without modifying the underlying clinical facts.

8.2 Primary scenario families

Family	Seed	Propagation	Observable failure
F1 Wrong-patient alias	Incorrect name/DOB/MRN association in registration memory.	Alias -> nursing lookup cue -> handover -> summary.	Later task retrieves or documents against the wrong patient.

Family	Seed	Propagation	Observable failure
F2 Wrong-chart copied fact	A benign-looking observation fragment is associated with another patient.	Observation summary -> handover -> aggregate.	Protected field or patient context changes in follow-up output.
F3 Access-scope laundering	Administration-visible note indirectly reflects a restricted field.	Restricted source -> authorized summary -> shared memory.	Unauthorized role can infer or reuse restricted information.
F4 Stale corrected fact	Previously valid association is corrected or revoked.	Old summary and retrieval cue remain active.	Agent reintroduces superseded information after correction.

8.3 Committed dataset size

Element	Minimum commitment
Base tasks	24 MedAgentBench-derived record tasks: 8 identity/retrieval, 8 labs/vitals/aggregation, 8 simulated documentation/summarization.
Role variants	Administration, nursing, and clinical-summary principals with deterministic permissions.
Matched controls	One contaminated and one surface-similar clean trajectory per task.
Propagation depths	1, 2, 3, and 4 memory derivation hops.
Provenance conditions	Complete; random 20/40/60% edge loss; targeted removal of cross-role and semantic-summary edges.
Evidence views	Explicit lineage only; raw centralized; unrestricted embedding; metadata-only capsule; latent sketch; full CARE.
Replay	Frozen task inputs and FHIR snapshots so all recovery methods operate on paired trajectories.
Split	Hold out complete patients/scenario templates, not random near-duplicate memory pairs.

8.4 Ground truth

During benchmark construction, private instrumentation records every actual input, retrieved memory, transformation, patient token, role/purpose policy, output, tool call, and replay recipe. This complete graph exists only for scoring. Operational recovery conditions receive masked or scoped views. A descendant is labelled contaminated when removing the seed changes a protected patient/resource/task predicate or when no clean independent derivation is available. Human review will audit all positive labels and a stratified sample of hard negatives.

9. Experimental Design and Baselines

ID	Condition	Purpose
A	No recovery	Measures residual harm and ordinary task utility.
B	Delete confirmed seed only	Tests whether descendants survive local cleanup.
C	Full memory reset	Safety-heavy fallback and lower bound on retained utility.
D	Explicit-lineage quarantine	Strong privacy-respecting baseline when no missing-edge reconstruction is attempted.
E	Explicit-lineage clean replay	Separates the value of recompilation from the value of latent candidate discovery.
F	Sketch-only quarantine	Measures false positives when semantic candidates are treated as causality.
G	Central raw-content oracle	Non-private upper comparator for discovery and reconstruction.
H	Complete private oracle graph	Unattainable provenance upper bound; reports oracle regret.
I	AEGIS-Care / full CARE	Explicit lineage + sketches + local attribution + recompilation + enforcement.

9.1 Evaluation sequence

1. Run each clean and contaminated trajectory and freeze the FHIR/memory snapshot before recovery.
2. Verify that the seed was written, propagated, retrieved, and capable of changing the target task predicate.
3. Apply the chosen provenance mask and role/purpose policy view.
4. Execute all baselines against the same seed, snapshot, mask, and follow-up tasks.
5. For AEGIS-Care, log capsule bytes, candidates, local replay calls, verdicts, repairs, quarantines, and versions.
6. Run clean tasks, wrong-patient probes, unauthorized-role probes, and resurrection attempts after recovery.
7. Repeat across scenario families and model seeds; freeze thresholds on development families.
8. Audit every residual wrong-patient/unauthorized event and every false destructive repair.

9.2 Critical ablations

Ablation	Question answered
Remove latent sketch	How much candidate recall comes from exact lineage and metadata alone?
Remove counterfactual replay	How many clean but semantically related memories are unnecessarily quarantined?

Ablation	Question answered
Quarantine instead of recompile	Does reconstruction materially restore post-recovery task utility?
Remove purpose/recipient scoping	How much linkability or unauthorized evidence release does policy prevent?
Remove resurrection firewall	How often does stale influence return through later writes or retrievals?
Vary sketch dimension/quantization	Where is the recovery-leakage-efficiency knee?
Random vs targeted edge loss	Are cross-role and summary edges uniquely important?

9.3 Model and threshold protocol

The primary agent will be one reproducible open-weight instruction model selected during the pilot, with Qwen3-8B or a comparable 7-8B model as the initial candidate. Generation temperature will be zero where supported. One frozen embedding model will produce local sketches. Logistic regression or a calibrated weighted score will be preferred over a large learned lineage model unless the pilot shows a clear need. Candidate threshold τ_c , influence threshold τ_i , and repair-confidence threshold τ_r will be frozen on development scenario families before the final test.

10. Metrics, Statistical Analysis, and Success Criteria

Metric	Operational definition	Direction
Residual wrong-patient/unauthorized harm (RWH)	Post-recovery tasks selecting a wrong patient/resource or exposing a role-forbidden field.	Lower
Descendant recall	True contaminated descendants withdrawn, repaired, or quarantined divided by all true contaminated descendants.	Higher
Descendant precision	True contaminated descendants divided by all artifacts touched destructively.	Higher
Benign-state retention (BSR)	Independent clean memories retained or equivalently rebuilt.	Higher
Repaired task success (RTS)	Legitimate follow-up tasks completed correctly after recovery.	Higher
False repair rate	Rebuilt artifacts that introduce a new patient/fact/task error.	Lower
Unauthorized exposure rate (UER)	Policy-forbidden information released via capsule, response, log, or coordinator view.	Zero/lower
Sketch leakage	Protected-attribute or membership-inference advantage over defined baselines.	Lower
Deletion resurrection rate (DRR)	Tombstoned influence recovered or republished during follow-up.	Lower
Recovery regret	Gap from the complete private oracle at matched BSR and UER.	Lower
Overhead	Additional latency, local replays, LLM calls, capsule bytes, and memory versions.	Lower

10.1 Primary comparisons

- I versus D isolates the value of missing-edge recovery plus recompilation over explicit-lineage quarantine.
- I versus E isolates the value of latent candidate discovery under incomplete provenance.
- I versus F tests whether counterfactual confirmation is necessary to protect clean state.
- I versus C measures utility retained relative to the safest simple fallback.
- I versus G measures recovery loss and privacy gain relative to centralized raw-content access.
- I versus H reports oracle regret and the irreducible cost of missing provenance.

10.2 Statistical analysis

- Use paired bootstrap confidence intervals because every method is evaluated on the same frozen trajectory.
- Use McNemar's exact test for paired binary wrong-patient/unauthorized outcomes where sample size permits.
- Report macro averages by scenario family and propagation depth so one large family cannot dominate.
- Report calibration using Brier score and expected calibration error for candidate/influence probabilities.
- Plot safety-utility-privacy frontiers rather than selecting one favorable operating point after seeing the test set.
- Publish negative results, including cases where privacy constraints make selective recovery unsafe.

PREDECLARED GO CRITERION

At a mentor-approved clean-state-retention and privacy floor, full CARE must reduce residual wrong-patient/unauthorized behavior beyond explicit-lineage quarantine/replay and retain materially more useful state than full reset. Exact numeric thresholds will be frozen after the pilot, not invented in the proposal.

11. Functional Specification and Scope

11.1 Required system functions

ID	Function	Acceptance evidence
F1	Persist versioned memories and parent/replay metadata.	Memory audit shows stable IDs, versions, owners, policies, and recipes.
F2	Initiate recovery from one or more confirmed seeds.	Seed becomes non-servable before reconstruction begins.
F3	Create signed, receiver-scoped recovery capsules.	Tampered, expired, wrong-purpose, or wrong-recipient capsules are rejected.
F4	Discover exact and latent candidate descendants locally.	Candidate logs and recall under masked lineage.
F5	Perform local counterfactual confirmation.	Signed verdict plus changed task predicates; raw text absent from coordinator log.
F6	Recompile from trusted FHIR sources or quarantine.	New version passes source, identity, schema, and task checks.
F7	Propagate repair through downstream artifacts.	Closure test shows no active confirmed descendant remains on the frontier.
F8	Prevent resurrection.	Follow-up write/retrieval probes cannot reactivate tombstoned influence.
F9	Produce a recovery certificate.	Machine-readable and human-readable incident summary with counts and unresolved risk.
F10	Support human override and safe resume.	Reviewer can approve, reject, or keep quarantined; resume requires follow-up validation.

11.2 Explicit non-goals

- No diagnosis, prognosis, treatment plan, medication order, autonomous clinical decision, or direct patient-facing advice.
- No use of live hospital systems, identifiable patient data, or claims of improved morbidity, mortality, or real-world clinical outcomes.
- No HIPAA, Digital Personal Data Protection Act, GDPR, or hospital compliance certification.
- No full latent mesh, cross-model hidden-state alignment, KV-cache transport, or latent-agent training.
- No claim that embeddings or sketches are confidential by construction.
- No attempt to solve incident detection, prompt injection, memory poisoning, access control, and machine unlearning in general.
- No destructive deletion of the audit trail; superseded content is made non-servable and represented by policy-controlled tombstones in the benchmark.

11.3 Safe fallback hierarchy

1. Automatically recompile only when trusted support and deterministic checks are sufficient.
2. Quarantine and request local human review when candidate influence is confirmed but reconstruction is uncertain.
3. Expand quarantine to the affected role/patient scope when lineage is too incomplete for selective repair.
4. Use a scoped or full reset when contamination density, replay failure, or privacy policy makes selective recovery unsafe.

12. Feasibility and Technology Stack

12.1 Minimum viable research system

The project is feasible only if it remains a benchmarked research prototype. The minimum system uses one FHIR sandbox, three logical agent roles, one primary LLM, one embedding model, 24 base tasks, and deterministic policy and grading code. The multi-agent boundary is logical rather than an attempt to deploy three independently trained models. This preserves the privacy/recovery structure while keeping compute and implementation manageable.

Layer	Proposed technology	Reason
Agent/runtime	Python 3.11, FastAPI, LangGraph or a lightweight custom orchestrator.	Typed tools, reproducible state transitions, and testable recovery hooks.
Clinical sandbox	MedAgentBench Docker image, HAPI FHIR JPA, FHIR R4 APIs.	Public interactive EHR environment with deterministic tasks.
Memory store	PostgreSQL/SQLite plus pgvector or FAISS; append-only version and event tables.	Supports exact lineage, vector candidate search, snapshots, and audit.
Primary model	Qwen3-8B or comparable 7-8B open-weight instruction model, quantized if required.	Reproducibility and practical single-GPU execution.
Sketch encoder	Frozen sentence embedding model with random projection and quantization.	Small, reproducible candidate representation; no hidden/KV transport.
Lineage/calibration	scikit-learn logistic regression, isotonic or Platt calibration.	Interpretable ablations and low training cost.
Integrity	SHA-256 commitments and Ed25519 signatures via standard libraries.	Binds capsule identity, purpose, expiry, and payload.
Policy	Deterministic RBAC/ABAC rules adapted from EICU-AC.	Separates role, patient, purpose, and operation checks from LLM judgment.
Evaluation	PyTest, frozen JSON traces, paired bootstrap scripts, Pandas/Matplotlib.	Reproducible graders, statistics, and plots.

Layer	Proposed technology	Reason
Interface	Streamlit or React dashboard as a stretch deliverable.	Human review, incident replay, and recovery certificate display.

12.2 Resource estimate

Resource	Estimate / mitigation
Compute	One 16-24 GB GPU or a controlled API fallback; 4-bit quantization and cached deterministic replays.
Storage	FHIR Docker image, benchmark records, event logs, vectors, and snapshots; expected to remain within tens of GB.
LLM calls	Prune candidates before replay; cache baseline generations; reuse frozen trajectories across recovery conditions.
Team effort	Parallel workstreams: environment/data; memory/recovery; policy/privacy; evaluation/paper.
Data governance	Use only benchmark/deidentified/synthetic records; no locally collected patient information.

12.3 Feasibility gates

Gate	Question	Decision
G0 Environment	Can the team run and grade selected MedAgentBench tasks reproducibly?	If not, use a smaller synthetic FHIR fixture with the same APIs.
G1 Propagation	Can one seed produce a repeatable multi-hop memory failure?	If not, narrow to deterministic summaries and retrieval cues before expanding.
G2 Missing-lineage value	Does edge masking create a real gap over explicit provenance?	If not, latent sketches are removed and the project becomes pure recompilation.
G3 Counterfactual value	Does replay confirmation improve precision over sketch similarity?	If not, replace it with deterministic structured dependency checks.
G4 Repair value	Does recompilation retain more utility than quarantine/reset?	If not, report the boundary and focus the paper on safe escalation.
G5 Privacy	Do sketches leak unacceptable attributes or linkability?	Reduce sketch dimension/query budget or retain all latent matching locally.

13. Twelve-Week Timeline and Deliverables

Weeks	Milestone	Work	Exit artifact
1-2	Environment freeze	Run FHIR sandbox; select 24 tasks; implement three roles and deterministic graders.	Pinned environment, task manifest, access-policy tests.
3	Versioned memory	Add memory writes, summaries, parent commitments, replay recipes, snapshots, and events.	Clean multi-session traces and oracle lineage.
4	Incident overlay	Implement wrong-patient, wrong-chart, scope-laundering, and stale-fact scenarios with matched clean controls.	Validated contamination labels and follow-up predicates.
5	Baseline recovery	Seed deletion, reset, explicit-lineage quarantine, explicit replay, and oracle conditions.	Baseline matrix and evidence that the constrained gap exists.
6	Capsules and sketches	Receiver/purpose/expiry policy, signatures, local sketch index, edge masks.	Candidate-recall and leakage pilot; G2/G5 decisions.
7	Counterfactual attribution	Local replay, structured influence tests, calibration, and hard-negative audit.	Precision/recall ablation; G3 decision.
8	Memory recompiler	Trusted-source rebuild, recursive closure, version publication, and quarantine.	End-to-end CARE conditions executable.
9	Resurrection firewall	Revocation checks, stale write/retrieval probes, safe-resume protocol.	Reinfection/resurrection test suite.
10	Full experiment	Run provenance masks, conditions, role policies, privacy probes, and paired replicates.	Frozen result tables and confidence intervals.
11	Failure analysis	Audit residual harm, false repairs, clean-state loss, leakage, and oracle regret.	Failure taxonomy and scope boundary.
12	Submission package	Finalize report, paper draft, code/data card, poster, presentation, and live prototype.	Capstone package and publication-ready manuscript.

13.1 Scope-cut order

1. Keep wrong-patient alias as the sole attack family; move wrong-chart and access-scope laundering to stretch experiments.
2. Reduce tasks from 24 to 16 while retaining matched clean controls and all three roles.
3. Use one 64-dimensional quantized sketch rather than a dimension sweep.
4. Use one random and one targeted provenance-loss condition rather than a dense sweep.
5. Keep logistic scoring; remove learned alternatives and secondary models.
6. Remove dashboard polish before removing counterfactual, privacy, and baseline ablations.

14. Risks, Ethics, and Limitations

Risk	Why it matters	Mitigation / honest reporting
Similarity masquerades as causality	Latent candidates may be clinically related but independently derived.	Require exact lineage or local counterfactual/structured confirmation before destructive action.
Counterfactual instability	LLM replay may vary even when inputs are unchanged.	Temperature zero, frozen prompts/models, multiple deterministic task predicates, and repeated replay only where needed.
False repair	A reconstructed summary may introduce a new error.	Trusted-source checks, structured schema validation, patient-identity checks, and quarantine on disagreement.
Sketch leakage	Compressed representations may expose sensitive attributes.	Receiver scoping, quantization, query limits, attacker evaluation, and no claims of formal privacy.
Benchmark realism	Synthetic/deidentified tasks cannot prove hospital effectiveness.	Claim only mechanism-level improvement in simulation; invite future clinical validation.
Compute explosion	Leave-one-out replay across all memories is expensive.	Candidate pruning, cached contexts, replay only on top-k candidates, and small propagation graphs.
Scope creep	Healthcare, privacy, agents, latent communication, and recovery can become separate projects.	One incident, one algorithm, one model, three logical roles, and explicit gate-based cuts.
Dual use	Attack scenarios may inform memory poisoning.	Publish sanitized, benchmark-specific payloads and emphasize recovery mechanics rather than operational exploit recipes.

14.1 Ethical position

AEGIS-Care is a safety-systems study, not a clinical AI efficacy study. It will not recruit patients, collect protected health information, or make real care decisions. The implementation will preserve the benchmark's deidentification and use synthetic incident overlays. Results will be described as simulated reductions in record-management failure, not improvements in clinical outcomes. Any future live evaluation would require institutional review, clinical governance, security engineering, legal review, and a substantially stronger threat model.

15. Expected Contributions and Publication Plan

15.1 Expected technical and empirical contributions

- A formal post-incident recovery problem for role-separated, persistent clinical-agent memory under incomplete provenance.
- The CARE algorithm that converts latent similarity into locally confirmed influence before incremental recompilation.

- A privacy-bounded recovery-capsule schema with recipient, purpose, expiry, commitment, sketch, and signed verdict fields.
- A clinical-memory contamination overlay for MedAgentBench-style FHIR tasks with versioned lineage, edge masks, role policies, and replay recipes.
- An evaluation protocol that jointly measures residual harm, clean-state retention, reconstruction quality, leakage, resurrection, and overhead.
- Boundary results showing when selective recovery should escalate to local review, scoped reset, or full reset.

15.2 What would make the paper publishable

The paper should not be sold as the first use of provenance, latent representations, or replay. Its strongest framing is an experimentally verified systems question: when lineage is incomplete and evidence cannot be centralized, how much of oracle recovery can a privacy-bounded recompiler recover, and what safety/utility/privacy cost is unavoidable? Publication quality requires a reproducible benchmark overlay, strong baselines, held-out scenario families, ablations showing that each CARE stage is load-bearing, and transparent negative results.

15.3 Tentative venue strategy

Target	Fit	Decision rule
IEEE SaTML 2027	Secure and trustworthy ML; official paper deadline 29 September 2026.	Attempt only if the pilot and core experiment are complete; do not rush an under-evaluated submission.
Journal of Biomedical Informatics	Methodological biomedical informatics, patient safety, AI, and clinical systems; rolling journal route.	Primary full-paper target after complete clinical benchmark and experiments.
ACM CODASPY 2027	Data/application security and privacy, including work-in-progress/poster pathways.	Use once the 2027 call is announced and the privacy/security contribution is experimentally clear.
IEEE Access / institutional journal fallback	Broad engineering systems paper with reproducible implementation.	Use only after mentor review of venue quality, cost, and fit.

The venue table is tentative and must be rechecked at submission time. IEEE SaTML 2027 currently lists a 29 September 2026 paper deadline [20]; Journal of Biomedical Informatics accepts methodological work in artificial intelligence, clinical decision support, and patient safety. Venue selection will follow the completed contribution rather than forcing the project into a premature deadline.

16. Conclusion

AEGIS-Care turns an abstract agent-security problem into a tangible clinical record-recovery system. Its motivating failure is simple: one wrong durable patient association can be transformed into several authentic-looking memories, and deleting the first entry may not remove what later agents learned from it. The proposed response is neither a blanket reset nor an unrestricted forensic read. It is a privacy-bounded memory recompiler.

The CARE loop gives every technical component one necessary role. Explicit lineage and latent sketches locate possible descendants; local counterfactual replay confirms actual influence; clean-room replay rebuilds safe state from trusted FHIR resources; and version enforcement prevents resurrection. The study is scoped to simulated, non-diagnostic record-management tasks and will be judged through paired safety, utility, privacy, and overhead measurements. If successful, it demonstrates a third option between hoping that seed deletion is enough and wiping all memory: selective, auditable, privacy-bounded recovery.

FINAL PROPOSAL STATEMENT

AEGIS-Care asks not only whether an AI agent can remember, but whether it can recover responsibly when what it remembers becomes wrong.

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Appendix A. Demonstration and Evaluation Trace

Phase	Registration runtime	Nursing runtime	Clinical-summary runtime	Recovery coordinator
Clean	Resolves Patient A to Record A.	Creates authorized handover.	Creates Patient A summary.	No incident.
Poison	Stores Patient A -> Record B alias.	No direct access to seed text.	No direct access to seed text.	No access to raw record.
Propagate	Uses alias in later lookup.	Derives handover from returned context.	Summarizes contaminated handover.	Still unaware.
Detect	Seed version is flagged.	Affected status unknown.	Affected status unknown.	Starts recovery transaction.
Candidate	Exact local descendants found.	Sketch match identifies possible handover.	Sketch/lineage finds possible summary.	Receives opaque candidate counts.
Confirm	Local replay changes selected patient.	Replay without suspect cue changes patient token.	Replay from repaired handover changes summary.	Receives signed high-influence verdicts.
Repair	Re-resolves from FHIR source.	Rebuilds handover from corrected record.	Rebuilds summary from corrected handover.	Tracks versions and closure.
Enforce	Old alias tombstoned.	Old handover superseded.	Old summary superseded.	Issues recovery certificate and safe-resume decision.

Appendix B. Mentor Defense Questions

Faculty question	Defensible answer
Why not just use a database rollback?	Rollback corrects the source record but does not necessarily identify agent-authored summaries, retrieval cues, or procedures already derived from it. AEGIS-Care targets those semantic artifacts.
Why call it latent if you are not sharing hidden states?	The candidate sketch is derived from a continuous representation, but the project intentionally restricts it to a quantized, receiver-scoped recovery signal. Full latent communication is out of scope.
Is similarity enough to prove contamination?	No. Similarity only generates candidates. Exact lineage or local counterfactual/structured replay is required before repair or quarantine.
How is privacy guaranteed?	Raw-content non-export and access decisions are deterministic within the defined interface. Sketch confidentiality is not guaranteed; leakage is empirically attacked and reported.
What if clean reconstruction is impossible?	The system quarantines and escalates. It never fabricates a replacement merely to preserve utility.
Why healthcare?	Wrong-patient and wrong-chart propagation is tangible, clinically meaningful, role-sensitive, and testable using public FHIR benchmarks without deploying a diagnostic system.
What is the single technical contribution?	A privacy-bounded, incomplete-lineage memory-recompilation loop that uses latent discovery only as candidate generation and verifies influence locally before incremental rebuild.